

The {marginaleffects} workflow

Every call makes three choices — quantity, grid, aggregation — and every call returns one data frame.

Fit the model

Estimate the coefficients

IN

turnout (2000 respondents)

```
f <- vote ~ age +  
  educate + income + race  
fit <- glm(f,  
  data = turnout,  
  family = binomial)
```

OUT

fit — the coefficient estimates $\hat{\beta}$ and their variance matrix.

Coefficients aren't interpretable on their own → compute quantities of interest (invariance property + delta method).

1 • Quantity

What do we compute?

a Scale • IN type =

DEFAULT probability scale. (comparisons() uses "response"; predictions() uses "invlink(link)" — same estimates, CI built on the link scale.)

OTHER "link" — linear predictor scale (rarely)

b Prediction or comparison?

IN which function you call

predictions() → expected value, E(y | X)

comparisons() → a comparison of two expected values, hi vs. lo

No default — you pick.

c What to compare (comparisons only) • IN variables =

DEFAULT every variable, one at a time:

numeric	observed → observed + 1	+1
factor	reference → each other level	white – others

OTHER focal variable age ; contrast label at right

list(age = 10)	age → age + 10	+10
list(age = c(18, 90))	18 → 90	90 – 18
list(age = "iqr")	25th → 75th %ile	Q3 – Q1
list(age = "2sd")	–1 SD → +1 SD	(x + sd) – (x – sd)

d How to compare • IN comparison =

DEFAULT	"difference"	hi – lo
OTHER	"ratio"	hi / lo
	"lift"	(hi – lo) / lo
	\(hi, lo) (hi / lo) – 1	custom function

OUT

A recipe for **one number per covariate row** — e.g., “Pr(vote) at age 90 minus Pr(vote) at age 18.”

2 • Grid

Where do we compute it?

IN newdata =

DEFAULT the observed data → one row per respondent (2000 rows)

OTHER built with datagrid(). Unspecified variables sit at their mean (mode for factors).

datagrid(type = "mean_or_mode")	the typical case	1 row
datagrid(age = 18:90)	vary age, fix the others	73 rows
datagrid(educate = c(12, 14, 16))	vary educate	3 rows
datagrid(race = unique)	vary race	2 rows

OUT

A table of covariate rows; the quantity is computed **once per row**.

Tip: inspect the output to see the grid you actually used.

3 • Aggregation

One estimate per row, or a summary?

IN which function you call

DEFAULT predictions() / comparisons()

Keep every row: one estimate per grid row, per contrast. Good for showing heterogeneity — plot them.

OTHER avg_predictions() / avg_comparisons()

Average the rows into one estimate; SE by the delta method (not mean() of the estimates' SEs).

Two named estimands (Hanmer & Kalkan 2013)

Average case = typical-case grid, then average (trivially, of 1 row)

```
avg_comparisons(fit,  
  newdata = datagrid(  
    type = "mean_or_mode"),  
  variables = ...)
```

Observed value = observed-data grid, then average the 2000

```
avg_comparisons(fit,  
  variables = ...)
```

OUT

The rows to report.

Output

Always a data frame

COLUMNS

identity	term, contrast (comparisons only)
estimate	estimate, std.error, conf.low, conf.high (95% by default)
grid	rowid, age, educate, income, race (row-level output only — dropped after avg_*())

HOW MANY ROWS?

predictions(fit)	2000
comparisons(fit)	8000
2000 × 4 contrasts	
comparisons(fit, variables = list(age = c(18, 90)))	2000
... newdata = datagrid(type = "mean_or_mode")	1
avg_comparisons(fit, variables = list(age = c(18, 90)))	1

rows = grid rows × contrasts, or one per contrast after avg_.

One call, all three choices

```
avg_comparisons(fit,  
  
  type = "response",  
  variables = list(age = c(18, 90)),  
  comparison = "difference",  
  newdata = datagrid(type = "mean_or_mode"))
```

3 • Aggregation

avg_ average the rows

1 • Quantity

comparisons a comparison, not a prediction

1 • Quantity

Fit fit the model

1 • Quantity

type scale (probability)

1 • Quantity

variables what to compare: age 18 → 90

1 • Quantity

comparison how to compare: hi – lo

2 • Grid

newdata where: the typical case

→ 1 row

term	contrast	estimate	conf.low	conf.high
age	90 – 18	0.299	0.236	0.361